B.E. Electrical Engineering Examination – 2023

SECOND YEAR, SECOND SEMESTER

ELECTRICAL MACHINES - II

PART - II COMPLETE SOLUTIONS

Comprehensive Academic Solution Manual

Instructions: Answer any three questions. Question no. 3 carries 18 marks. All sub-questions from Part-II are solved comprehensively below.

Contents

Question 1(i)

Problem Statement:
With the help of B-H characteristics of magnetic core explain how harmonics are introduced in a transformer.

Thought Process

To answer this analytically: 1. Introduce the non-linear relationship between flux (Φ) and magnetizing current (i_m) defined by the hysteresis (B-H) loop. 2. Discuss the two operational scenarios: - Scenario A: If the applied voltage is perfectly sinusoidal, the flux must be sinusoidal. Due to the non-linear B-H curve, the required magnetizing current must become highly peaked (containing prominent 3rd harmonic components). - Scenario B: If the current is forced to be purely sinusoidal, the non-linear saturation causes the flux to become flat-topped (which also introduces 3rd harmonics in the flux, leading to peaked induced voltages). 3. Draw a graphical projection diagram showing a sinusoidal flux translating through a B-H curve to yield a peaked, non-sinusoidal magnetizing current.

Answer:
The introduction of harmonics in a transformer is fundamentally rooted in the non-linear magnetic properties of its silicon-steel core, specifically governed by its **B-H (hysteresis) characteristics** and magnetic saturation.

1. The Fundamental Cause: Core Non-Linearity

According to Faraday's Law, if a transformer is connected to a purely sinusoidal infinite bus voltage ($v = V_m \cos \omega t$), the mutual core flux (Φ) must also be purely sinusoidal to balance the applied voltage:

$$e = N\frac{d\Phi}{dt} \implies \Phi(t) = \Phi_m \sin \omega t \tag{1}$$

However, the relationship between the required magnetizing current (i_m) and the core flux (Φ) is defined by the magnetic core's hysteresis loop. Because the B-H curve is **non-linear and exhibits saturation**, a linearly increasing current does not produce a linearly increasing flux.

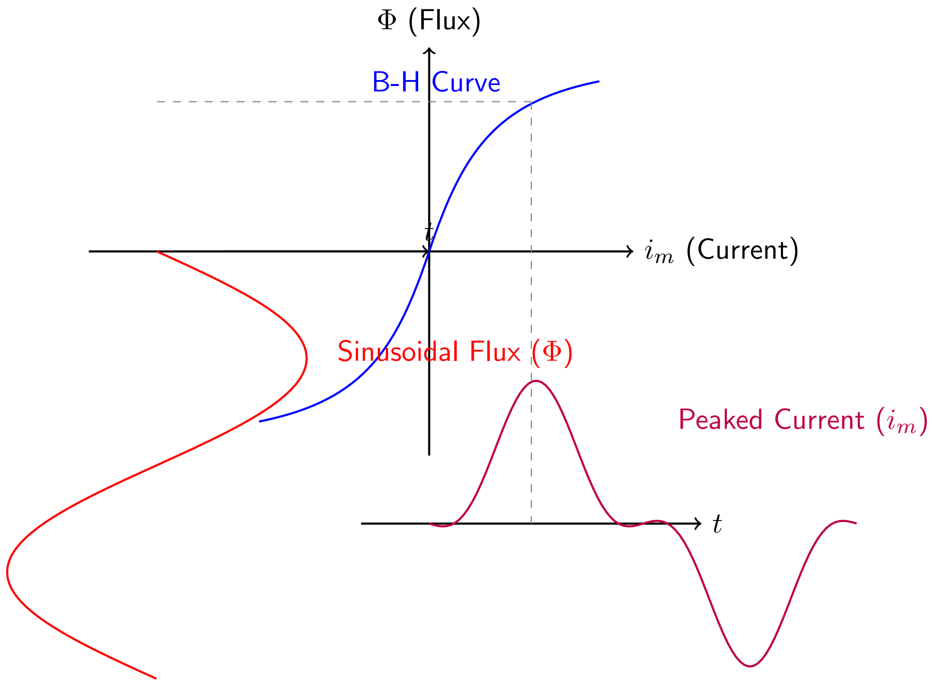
2. Introduction of the Third Harmonic

To sustain a perfect sinusoidal flux near the saturation knee-point of the core, the transformer requires disproportionately larger amounts of current at the peaks of the flux wave. Consequently, the magnetizing current i_m becomes highly distorted and **peaked**. When this non-sinusoidal peaked current waveform is analyzed using Fourier series, it is found to consist of:

- A fundamental component ($I_1 \sin \omega t$)
- A prominent **third harmonic component** ($I_3 \sin 3\omega t$)
- Smaller odd harmonics ($5^{\text{th}}, 7^{\text{th}}, \dots$)

Therefore, harmonics are natively introduced because the transformer *demands* harmonic currents from the supply to maintain normal sinusoidal flux operation. If the supply network configuration prevents these third harmonic currents from flowing (e.g., in a 3-wire star connection), the core flux is forced to become flat-topped, introducing harmonic distortions directly into the induced secondary voltages.

Graphical Projection of Harmonics via B-H Curve


Φ (Flux)
B-H Curve
t
i_m (Current)
Sinusoidal Flux (Φ)
Peaked Current (i_m)
t

Question 1(ii)

Problem Statement:
Show that Delta winding is better in harmonics related issues in a three phase transformer?

Thought Process

Key analytical points to address: 1. Explain how a closed Delta (Δ) loop interacts with co-phasor triplen harmonic currents. 2. Show mathematically that triplen harmonics sum to zero at the line terminals but circulate continuously within the closed mesh loop. 3. Detail how this circulating current provides the necessary internal magnetomotive force (MMF) to restore sinusoidal core flux, effectively mitigating harmonic voltage distortion without exporting harmonic currents to the external grid.

Answer:
A delta (Δ) connected winding (whether primary, secondary, or tertiary) serves as an inherent, self-contained solution to harmonic distortion in three-phase transformers. Its superiority is due to its closed-loop electrical topology.

1. Triplen Harmonics are Co-Phasor

In a balanced three-phase system, the fundamental currents are displaced by 120°. However, for the third harmonic (and its odd multiples: $9^{\text{th}}, 15^{\text{th}}$, collectively called triplen harmonics), the phase displacement becomes $3 \times 120° = 360°$.

$$i_{a3} = I_{3m} \sin(3\omega t) \tag{2}$$
$$i_{b3} = I_{3m} \sin(3(\omega t - 120°)) = I_{3m} \sin(3\omega t - 360°) = I_{3m} \sin(3\omega t) \tag{3}$$
$$i_{c3} = I_{3m} \sin(3(\omega t + 120°)) = I_{3m} \sin(3\omega t + 360°) = I_{3m} \sin(3\omega t) \tag{4}$$

Thus, triplen harmonic currents are exactly **in phase** with one another in all three windings.

2. Trapping Harmonics in the Delta Loop

In a delta connection, the line current (I_{line}) leaving any node is the phasor difference between the adjacent phase currents:

$$I_{\text{line, 3}} = I_{\text{phase, a3}} - I_{\text{phase, b3}} \tag{5}$$

Because the third harmonic phase currents are identical and in-phase ($I_{a3} = I_{b3}$), the resulting line current is exactly **zero**:

$$I_{\text{line, 3}} = 0 \tag{6}$$

Consequently, triplen harmonic currents **cannot flow out into the external transmission lines**, preventing telephonic interference and power grid pollution. Instead, because they are co-phasor, they sum constructively and **circulate continuously around the closed mesh loop** of the delta winding.

3. Mitigation of Flux Distortion (MMF Provision)

Due to core saturation, the transformer requires a 3rd harmonic magnetizing current to maintain a sinusoidal flux.

- If a flat-topped flux attempts to develop, it induces a 3rd harmonic EMF (E_3) within the delta winding.
- This E_3 drives the circulating 3rd harmonic current (I_{3c}) around the low-impedance delta loop.

- This circulating current acts as an internal magnetomotive force (MMF) that naturally supplies the missing harmonic excitation needed by the core.
- As a result, the core flux is restored to a pure sinusoid, guaranteeing that the induced secondary phase voltages are perfectly sinusoidal and free from dangerous peaking.

Because it inherently traps harmonics and stabilizes the flux, a delta winding is highly superior for managing harmonic-related issues.

Question 1(iii)

Problem Statement:
What is oscillating neutral problem? - Explain with proper diagram.

Thought Process

Steps for a comprehensive response: 1. Identify the connection: Occurs in ungrounded Star-Star (Y-y) configurations. 2. Explain the cause: Suppression of 3rd harmonic currents forces the flux to become flat-topped. 3. Explain the effect: Flat-topped flux induces highly peaked 3rd harmonic phase voltages. 4. Define the problem: Since 3rd harmonic voltages are co-phasor, they shift the physical potential of the star neutral point relative to the ground at a frequency of 3× the fundamental (150 Hz). 5. Provide a phasor diagram demonstrating the neutral shift.

Answer:
The **oscillating neutral problem** (also known as the floating neutral phenomenon) occurs primarily in **Star-Star (Yy)** connected transformers lacking a primary neutral return wire or a delta stabilizing winding.

Mechanism of the Problem

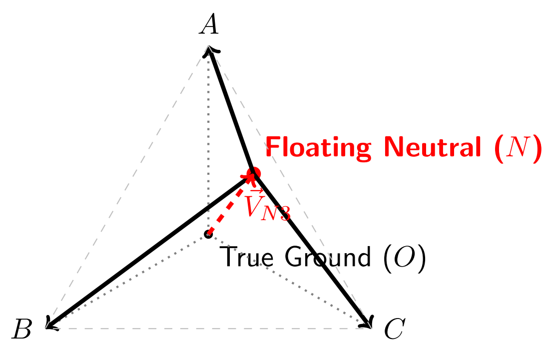
1. **Suppression of 3rd Harmonic Current:** In a 3-wire star connection without a neutral, Kirchhoff's Current Law dictates that the sum of the line currents entering the star point must be zero ($i_a + i_b + i_c = 0$). Since 3rd harmonic currents are co-phasor ($i_{a3} = i_{b3} = i_{c3}$), their sum is $3i_{a3} = 0$, meaning they are completely suppressed.

2. **Peaking of Phase Voltages:** Deprived of the required 3rd harmonic magnetizing current, the core flux becomes severely **flat-topped**. According to Faraday's Law ($e = -Nd\Phi/dt$), the derivative of a flat-topped wave produces an extremely peaked induced voltage waveform containing a massive 3rd harmonic voltage component (e_3).

3. **Oscillation of the Neutral Point:** In a balanced system, the fundamental frequency voltages sum to zero at the neutral point, anchoring its potential to ground (0V). However, the induced 3rd harmonic voltages in the three phases are identical and in-phase. Therefore, they do not cancel out at the star point. Instead, they act additively between the system ground and the physical neutral point of the transformer.

This causes the entire neutral point to physically oscillate at three times the fundamental frequency (e.g., 150 Hz on a 50 Hz system) with respect to the true ground. This instability leads to severe phase-voltage unbalances, high dielectric stress on the insulation, and potential failure of equipment connected between phase and neutral.

Phasor Diagram of the Oscillating Neutral



Question 2(i)

Problem Statement:
Explain the importance of vector groups for three phase transformers.

Thought Process

Analytical requirement: Explain why vector grouping is a critical engineering standard. The primary reason is to ensure the safe and stable parallel operation of transformers. Mention phase displacement, synchronization, and the prevention of catastrophic short-circuit circulating currents.

Answer:
Vector grouping in three-phase transformers is an internationally standardized method (using a clock-face convention) to classify the primary and secondary winding configurations (Star, Delta, Zig-zag) and the inherent phase angle displacement between their respective line voltages.

The critical importance of vector groups lies in the following areas:

1. **Parallel Operation Safety:** The single most important reason for identifying vector groups is to determine if two transformers can be safely operated in parallel. For successful paralleling, the secondary line voltages of both units must have **exactly zero relative phase displacement**. If two transformers belonging to incompatible vector groups (e.g., a Dy1 with a $-30°$ shift and a Yy0 with a $0°$ shift) are paralleled, the angular voltage difference between their secondary terminals will drive a massive, catastrophic circulating short-circuit current through the low-impedance windings, instantly destroying the transformers.

2. **Determining Connection Modifications:** Knowing the vector group allows engineers to deduce if connection modifications can make incompatible units parallelable. For instance, a $+30°$ transformer (Group 4) can be paralleled with a $-30°$ transformer (Group 3) by deliberately reversing specific primary and secondary phase sequences.

3. **Harmonic and Grounding Configuration:** The vector group designation immediately informs system protection engineers whether a delta winding is present (to trap 3rd harmonics) and whether a neutral point is available for solid or impedance grounding (indicated by the 'n' in groups like Dyn11).

Question 2(ii)

Problem Statement:
Draw the connection diagram and phasor diagram of the following connections. a) Dd6, b) Dz0 and c) Dy11

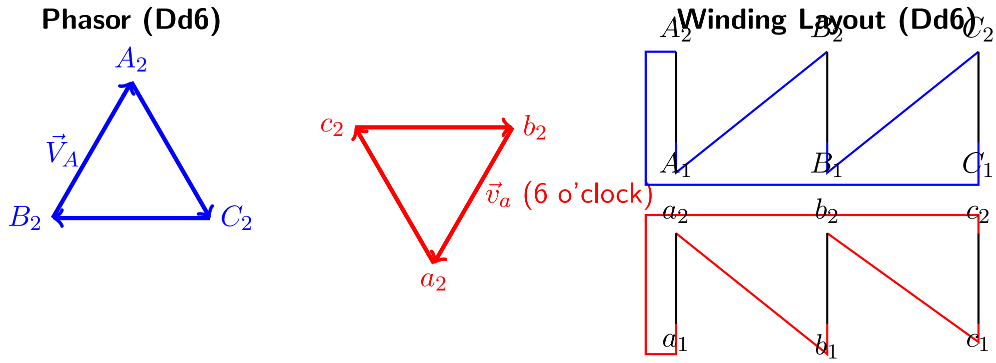
Thought Process

Requirements: 1. **Dd6**: Delta-Delta, $180°$ phase displacement (6 o'clock). 2. **Dz0**: Delta-Zigzag star, $0°$ phase displacement (12 o'clock). 3. **Dy11**: Delta-Star, $+30°$ leading phase displacement (11 o'clock). Draw both the winding schematic and the clock phasor diagrams for each.

Answer:
In standard clock convention, the High Voltage (HV) primary phase voltage acts as the reference minute hand fixed at 12 o'clock. The Low Voltage (LV) terminal acts as the hour hand.

a) Configuration: Dd6 (Delta-Delta, $180°$ Phase Displacement)

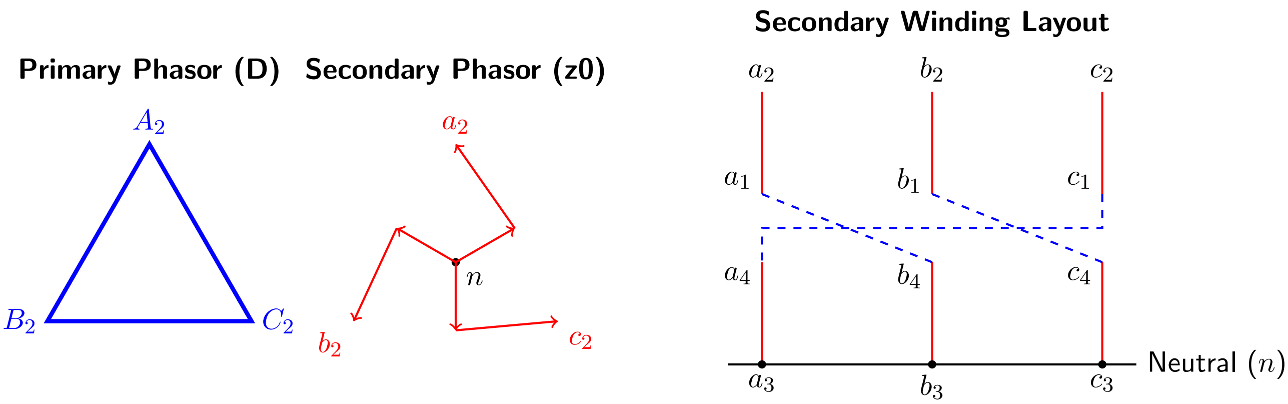
Both windings are delta connected, but the sequence of connections on the secondary is reversed relative to the primary, flipping the phasor $180°$ to the 6 o'clock position.



b) Configuration: Dz0 (Delta-Zigzag, $0°$ Phase Displacement)

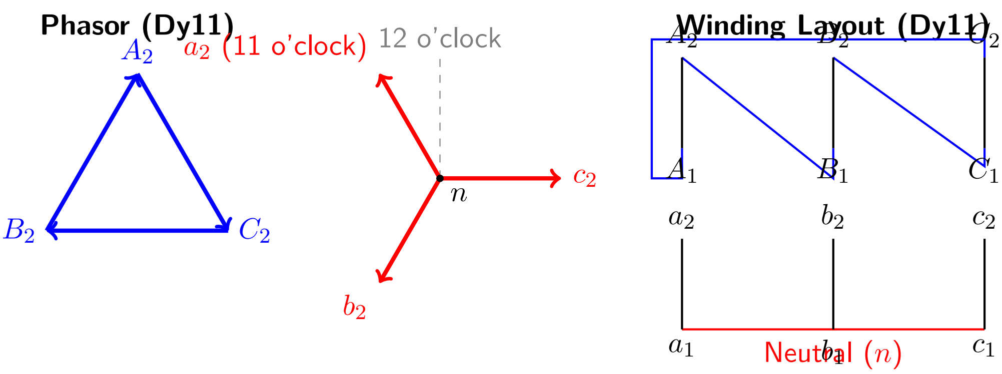
The primary is Delta. The secondary is split and cross-connected in a zigzag pattern carefully chosen so that the vector addition aligns exactly at 12 o'clock ($0°$).

Dz0 Transformer Connection and Phasor Diagram



c) Configuration: Dy11 (Delta-Star, $+30°$ Phase Displacement)

The primary is Delta. The secondary is Star, connected such that the secondary line voltages lead the primary line voltages by $30°$, putting the a_2 phasor at 11 o'clock.



Question 2(iii)

Problem Statement:
What will happen if a Dy1 connected transformer is paralleled with a Dy11 connected transformer? Show how they can be connected properly with connection modification.

Thought Process

Analytical steps: 1. Identify the clash: Dy1 is $-30°$ (Group 3), Dy11 is $+30°$ (Group 4). The phase difference is $60°$. 2. Consequence: The voltage difference $\Delta V = V \sin(60°/2) = V$. This creates a massive internal short-circuit current causing failure. Direct paralleling is forbidden. 3. Modification: Interchanging two primary input supply lines reverses the internal phase rotation sequence. Swapping B and C makes the $+30°$ lead become a $-30°$ lag. Finally, cross-swap the corresponding secondary lines to restore external a-b-c rotation to the bus.

Answer:

1. Consequences of Direct Paralleling

If a Dy1 transformer (Phase shift: $-30°$) is connected directly in parallel with a Dy11 transformer (Phase shift: $+30°$) to the same primary supply, there will be a severe angular discrepancy between their corresponding secondary terminals.

$$\text{Phase Difference } (\alpha) = +30° - (-30°) = 60° \tag{7}$$

Because the vectors are separated by $60°$, the resultant voltage difference ($\Delta\vec{V}$) across the paralleling switch contacts is:

$$|\Delta\vec{V}| = 2 \cdot V_2 \cdot \sin\left(\frac{60°}{2}\right) = V_2 \tag{8}$$

This means the voltage driving current between the two transformers is equal to the full secondary phase voltage. Because the internal leakage impedance is small, this acts as a **dead line-to-line short circuit**. A massive, destructive circulating current will instantly burn out the windings. Direct paralleling is impossible.

2. Connection Modification Strategy

A Group 4 (Dy11) transformer can be externally modified to operate identically to a Group 3 (Dy1) transformer by performing a dual phase-sequence reversal:

1. **Primary Swap:** Interchange two of the three-phase supply lines connecting to the primary terminals of the Dy11 transformer. Keep Phase A on A_2. Connect supply line B to C_2, and supply line C to B_2.

2. **Resulting Internal Inversion:** By reversing the primary phase sequence (from A-B-C to A-C-B), the internal rotating magnetic field is reversed. The original $+30°$ phase lead flips to a **$-30°$ phase lag**. The transformer is now internally operating as a Dy1 unit.

3. **Secondary Output Swap:** Because the primary sequence was reversed, the secondary line voltage sequence is also reversed ($a - c - b$). Before connecting the secondary cables to the load busbar, swap the b and c output cables. Connect terminal a_2 to bus a, terminal c_2 to bus b, and terminal b_2 to bus c.

With this modification, the secondary voltage phasors of both transformers will perfectly coincide ($\Delta\vec{V} = 0$), allowing safe and successful parallel load sharing.

Question 3(i)

Problem Statement:
Is it possible to connect two single phase transformers, with one having a center tap and the other with no tap, in a Scott-connected transformer? Discuss with proper diagrams.

Thought Process

Reasoning: 1. The Scott connection requires a Main transformer (with a 50% center tap) and a Teaser transformer (which theoretically requires an 86.62. The user asks what happens if the teaser has **no tap** (1003. If we connect the 1004. Since it uses 1005. Conclusion: It *is* possible to connect them structurally to get a 90-degree phase shift, but it **will not produce a balanced 2-phase supply** unless the secondary turns of the teaser are artificially altered or a booster is used.

Answer:
Yes, it is physically possible to wire them in a Scott configuration to achieve a 90-degree (two-phase) displacement, but **it will fail to produce a balanced two-phase voltage output** if the secondary windings of both transformers are identical.

Analysis of the Imbalance

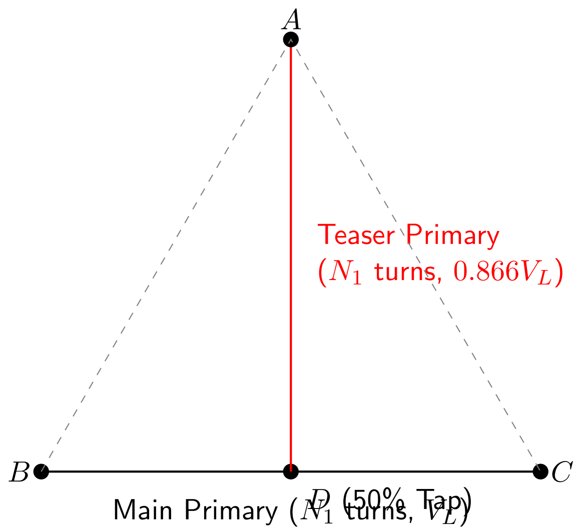
Let the 3-phase input line voltage be V_L. Let both single-phase transformers have an identical total primary turns count (N_1) and secondary turns count (N_2).

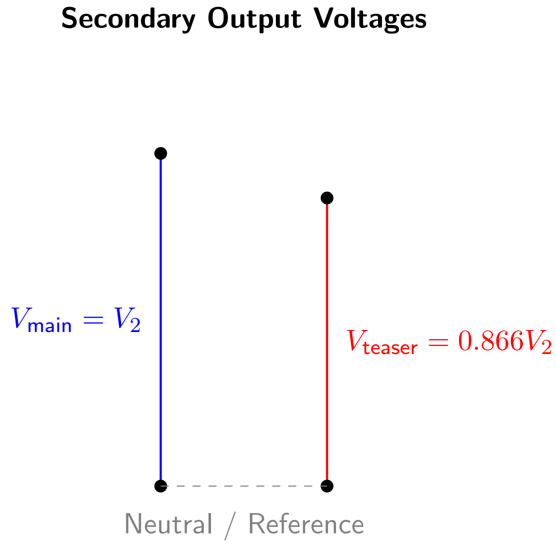
1. **Main Transformer:** The primary is connected across lines B and C and utilizes its 50% center tap (node D). The voltage across it is exactly V_L. The output voltage on the secondary is:
$$V_{\text{main\_sec}} = V_L \times \left(\frac{N_2}{N_1}\right) \tag{9}$$
2. **Teaser Transformer (No Tap):** The primary is connected between line A and the center tap D. From phasor geometry, the voltage applied to this primary winding is exactly altitude of the equilateral triangle:
$$V_{AD} = \frac{\sqrt{3}}{2}V_L = 0.866V_L \tag{10}$$
Because the teaser has **no tap**, the entire N_1 turns must be used to carry this reduced voltage. Consequently, the output voltage on its secondary is:
$$V_{\text{teaser\_sec}} = 0.866V_L \times \left(\frac{N_2}{N_1}\right) \tag{11}$$

Conclusion

Comparing the two outputs, $V_{\text{teaser\_sec}}$ is only **86.6%** of $V_{\text{main\_sec}}$. While the phase angle between the two secondary voltages is perfectly orthogonal (90°), the magnitude is significantly unbalanced. To fix this and provide a true balanced 2-phase supply using an untapped teaser, one would have to use an external auto-transformer to step up the voltage V_{AD} back to full V_L before feeding it to the teaser, or fundamentally alter the N_2 turn count of the teaser winding.

Scott Connection with Untapped Teaser Diagram


A
Teaser Primary
(N_1 turns, $0.866V_L$)
B
C
D (50% Tap)
Main Primary (N_1 turns, V_L)


Secondary Output Voltages
$V_{\text{main}} = V_2$
$V_{\text{teaser}} = 0.866V_2$
Neutral / Reference

Question 3(ii)

Problem Statement:
In Scott-connected transformers, teaser transformer supplies 0.75 lagging power factor load of 20 kW at 220 V and main transformer supplies 0.85 power factor leading load of 40 kW at 220 V, from a three phase input line voltage of 3300V. Determine the input 3-phase line currents. Neglect magnetizing currents and the leakage impedance drops. Draw input current phasors computed for above along with a Scott-connection diagram.

Thought Process

Step-by-step mathematical plan: 1. Transformation ratio: $v = N_P/N_S = 3300/220 = 15$. 2. Secondary load currents (Complex form): - Teaser load (P_T = 20kW, 0.75 lag). $\vec{V}_{T2}$ is reference at 90°. I_{T2} lags $\vec{V}_{T2}$ by 41.41°. - Main load (P_M = 40kW, 0.85 lead). $\vec{V}_{M2}$ is reference at 0°. I_{M2} leads $\vec{V}_{M2}$ by 31.79°. 3. Map to primary currents $\vec{I}_A, \vec{I}_B, \vec{I}_C$ via MMF balancing.

Answer:

1. Load Current Calculations on Secondary Side

The transformer ratio is $v = \frac{3300}{220} = 15$. Let the main secondary voltage act as the horizontal reference phasor ($\vec{V}_{M2} = 220\angle 0°$). In a Scott connection, the teaser secondary voltage is in quadrature ($\vec{V}_{T2} = 220\angle 90°$).

Teaser Secondary Current ($\vec{I}_{T2}$): The teaser supplies a 20 kW load at 0.75 lagging power factor.

$$I_{T2} = \frac{20,000}{220 \times 0.75} = 121.212 \text{ A} \tag{12}$$

Since it is lagging, the current phasor is behind the teaser voltage by $\phi_T = \arccos(0.75) = 41.41°$.

$$\vec{I}_{T2} = 121.212\angle(90° - 41.41°) = 121.212\angle 48.59° \text{ A} \tag{13}$$

Main Secondary Current ($\vec{I}_{M2}$): The main transformer supplies a 40 kW load at 0.85 leading power factor.

$$I_{M2} = \frac{40,000}{220 \times 0.85} = 213.904 \text{ A} \tag{14}$$

Since it is leading, the current phasor is ahead of the main voltage by $\phi_M = \arccos(0.85) = 31.79°$.

$$\vec{I}_{M2} = 213.904\angle(0° + 31.79°) = 213.904\angle 31.79° \text{ A} \tag{15}$$

2. Derivation of Primary Line Currents

Line Current A ($\vec{I}_A$): This current solely feeds the teaser primary. Due to the 86.6% tap, the MMF balance dictates:

$$\vec{I}_A = \frac{2}{\sqrt{3}} \cdot \frac{1}{v} \cdot \vec{I}_{T2} = \frac{2}{\sqrt{3}} \cdot \frac{1}{15} \cdot 121.212\angle 48.59° = 9.331\angle 48.59° \text{ A} \tag{16}$$

$$\vec{I}_A = 9.331(\cos 48.59° + j \sin 48.59°) = 6.172 + j7.000 \text{ A} \tag{17}$$

Line Currents B and C ($\vec{I}_B, \vec{I}_C$): The primary current across the main winding ($\vec{I}_{M1}$) is translated directly by the base turns ratio v:

$$\vec{I}_{M1} = \frac{\vec{I}_{M2}}{v} = \frac{213.904\angle 31.79^\circ}{15} = 14.260\angle 31.79^\circ \text{ A} \tag{18}$$

$$\vec{I}_{M1} = 14.260(\cos 31.79^\circ + j\sin 31.79^\circ) = 12.121 + j7.513 \text{ A} \tag{19}$$

The return current from the teaser splits equally at the center tap. Applying Kirchhoff's laws at the nodes:

$$\frac{\vec{I}_A}{2} = \frac{6.172 + j7.000}{2} = 3.086 + j3.500 \text{ A} \tag{20}$$

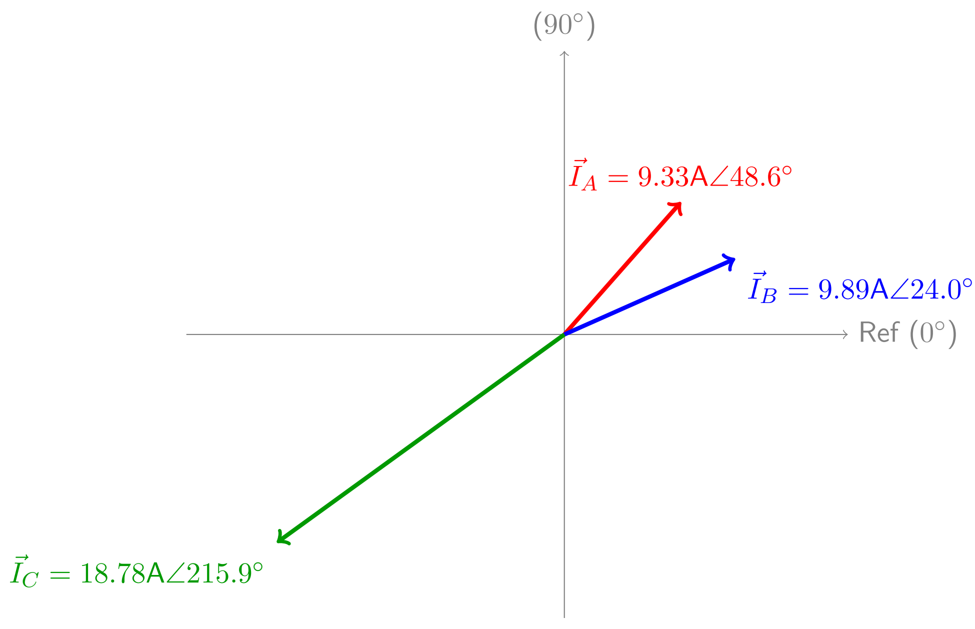
$$\vec{I}_B = \vec{I}_{M1} - \frac{\vec{I}_A}{2} = (12.121 + j7.513) - (3.086 + j3.500) = \mathbf{9.035 + j4.013} \text{ A} \tag{21}$$

$$|\vec{I}_B| = \sqrt{9.035^2 + 4.013^2} = \mathbf{9.886} \text{ A}, \quad \angle\vec{I}_B = \arctan\left(\frac{4.013}{9.035}\right) = \mathbf{23.95}^\circ \tag{22}$$

$$\vec{I}_C = -\vec{I}_{M1} - \frac{\vec{I}_A}{2} = (-12.121 - j7.513) - (3.086 + j3.500) = \mathbf{-15.207 - j11.013} \text{ A} \tag{23}$$

$$|\vec{I}_C| = \sqrt{(-15.207)^2 + (-11.013)^2} = \mathbf{18.775} \text{ A}, \quad \angle\vec{I}_C = 180^\circ + 35.91^\circ = \mathbf{215.91}^\circ \tag{24}$$

3. Phasor Diagram



Question 4(i)

Problem Statement:
Discuss the placement of taps in a three phase transformer.

Thought Process

Discuss why taps are placed on the High Voltage (HV) winding and specifically at its geometric center (middle). Reasons include current minimization, resolution of voltage steps, mechanical symmetry during faults, and protection against impulse surges.

Answer:
In three-phase power transformers, voltage regulating taps are almost universally placed on the **High Voltage (HV) winding**, and specifically located at the **geometric middle** of the winding structure. The technical justifications for this standard practice are:

1. **Current Interruption Capabilities:** The HV winding carries significantly less current than the low-voltage (LV) winding. Placing the tap changer on the HV side minimizes the current that the physical contacts must interrupt during tap changing, drastically reducing arc energy and contact wear, and allowing for a smaller, cheaper switch mechanism.

2. **Fine Voltage Resolution:** The HV winding contains a much larger number of turns. This allows for precise fractional turn adjustments. For instance, obtaining a $\pm 1.25\%$ voltage step is easily achievable on a multi-thousand turn HV winding, but might require a physical fraction of a turn on a low-turn LV winding, which is impossible.

3. **Maintenance of Magnetic Symmetry (Axial Forces):** During an external short-circuit fault, transformers experience massive electrodynamic forces. If taps were placed at the extreme end of the winding, cutting out turns would create severe vertical magnetic asymmetry between the primary and secondary. This creates massive **axial mechanical forces** capable of destroying the transformer. Placing taps at the center allows turns to be cut symmetrically from the middle, maintaining balanced magnetic centers.

4. **Protection from Surge Voltages:** Lightning and switching impulse surges distribute highly non-linearly across a winding, causing extreme dielectric stress concentrated primarily on the first few turns near the line terminals. Placing the tap changer mechanism in the middle of the winding shields its sensitive contacts and insulation from these peak transient voltages.

Question 4(ii)

Problem Statement:
Component wise what is the basic difference of an Off-load and an ON-load tap changer?

Thought Process

Key distinctions: Off-load tap changers are simple mechanical switches that require the transformer to be completely de-energized. ON-load tap changers (OLTC) are complex systems that must switch taps while current flows. They require specific components: Transition impedances (resistors/reactors) and a high-speed Diverter Switch mechanism.

Answer:
The primary difference between these mechanisms lies in their ability to handle live current interruption and prevent tap-to-tap short circuits.

Off-Load Tap Changer Components: An off-load (or de-energized) tap changer is structurally very simple. It consists solely of a **mechanical selector switch**. Because it is strictly operated only when the transformer is disconnected from the high-voltage supply, it does not require arc-quenching mechanisms, high-speed springs, or current-limiting devices. The contacts simply break their physical connection and rotate to the next tap.

On-Load Tap Changer (OLTC) Components: An OLTC must change taps while the transformer is actively supplying power to a load. It requires a sophisticated suite of components to ensure the circuit is never fully opened (which would cause massive arcing) and adjacent taps are never directly shorted. The unique components include:

- **Transition Impedance (Resistor or Reactor):** Temporary current-limiting devices inserted into the circuit during the brief moment when the switch bridges two adjacent taps, preventing a destructive short-circuit circulating current.
- **Tap Selector Switch:** A mechanical arm that pre-selects the upcoming tap under a no-load condition before the main current transfer occurs.
- **Diverter Switch (Arising Contact System):** A high-speed, spring-loaded switch mechanism (often housed in a separate oil compartment) that rapidly diverts the live load current from the old tap through the transition impedances to the new tap, effectively quenching arcs within milliseconds.

Question 4(iii)

Problem Statement:
Describe the operation of a reactor type tap changers.

Thought Process

Outline the operation: 1. Explain the use of a center-tapped preventive auto-transformer (reactor). 2. Differentiate from the resistor type: Reactor types can operate continuously in the "bridging" position. 3. Detail the sequence: Tap 1 operation, Bridging operation (Tap 1 + Tap 2), Tap 2 operation.

Answer:
A **reactor-type On-Load Tap Changer (OLTC)** uses a continuously rated, center-tapped preventive auto-transformer (reactor) as its transition impedance. Unlike resistor-type OLTCs—which use high-speed diverter switches to insert resistors for only milliseconds to prevent overheating—reactor types are highly robust and are actively designed to **operate continuously** in the intermediate bridging position, effectively doubling the number of available voltage steps.

Operational Sequence

Let us trace the operation as the system steps up from Tap 1 to Tap 2. The reactor has two input terminals (X and Y) connected to selector switches, and a center-tap output going to the load.

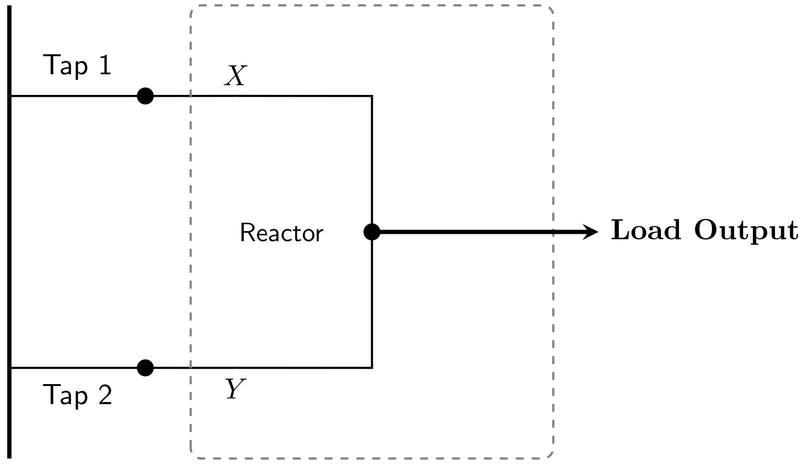
1. **Steady State (Tap 1 Only):** Both selector switch arms (X and Y) are connected to Tap 1. The load current splits equally through the two halves of the reactor. Because the currents flow in opposite directions toward the center tap, their magnetic fluxes cancel out. The reactor offers almost zero impedance (only ohmic resistance) to the load current.

2. **Pre-transition:** Selector arm Y opens from Tap 1. The entire load current momentarily flows through arm X and one half of the reactor. The reactor now acts as a series choke, slightly dropping the voltage, but arcing is minimized by the reactor's high inductance.

3. **Bridging Position (Tap 1 + Tap 2):** Selector arm Y connects to Tap 2, while arm X remains on Tap 1.

   - The reactor bridges the two adjacent taps. The voltage difference between Tap 1 and Tap 2 drives a circulating current through the full length of the reactor coil. Its high inductive reactance safely limits this circulating current.
   - Simultaneously, the load current splits through both halves to the center tap. The output voltage delivered to the load is exactly the **average** of Tap 1 and Tap 2.
   - The OLTC can remain indefinitely in this bridging position, serving as an operational half-step voltage.

4. **Final Transition (Tap 2 Only):** Selector arm X opens from Tap 1, momentarily placing all load current on arm Y. Arm X then moves and connects to Tap 2 alongside arm Y. The circulating loop is broken, flux cancellation is restored, and the transformer operates fully on Tap 2.


Tap 1
X
Reactor
Load Output
Tap 2
Y

Question 5(i)

Problem Statement:
Write Short notes on: Full wave and chopped wave impulse voltage tests on a transformer for detection of faults in the winding.

Thought Process

Key details: 1. Purpose: Simulating lightning or switching surges (1.2/50 μs wave). 2. Full-Wave Test: Application of unchopped surge. Faults detected by comparing neutral current oscillograms against a baseline. 3. Chopped-Wave Test: Simulates insulator flashover by chopping the wave at 2-6 μs. Tests inter-turn insulation against severe dv/dt.

Answer:
To verify that a transformer's insulation system can withstand high-voltage transient surges caused by lightning strikes and network switching, units are subjected to standard high-voltage impulse tests. The standard unchopped testing waveform has a fast front time of 1.2 μs and a long tail time of 50 μs (the 1.2/50 μs wave).

1. Full Wave Impulse Voltage Test

The full wave test applies the complete, unchopped 1.2/50 μs surge wave at a specified Basic Insulation Level (BIL) voltage directly to the transformer line terminals.

- **Fault Detection Mechanism:** Highly sensitive high-speed Cathode Ray Oscilloscopes (CROs) record the transient voltage wave shape and the resulting current waveform flowing out of the grounded neutral point.
- **Analysis Method:** Before applying the full test voltage, a low-voltage calibration shot (typically at 50% BIL) is applied to establish a baseline waveform signature. If an insulation failure occurs during the full-scale test (such as a breakdown between adjacent turns), the sudden change in internal impedance alters the wave profile. **Any deviation or high-frequency ringing in the neutral current oscillogram compared to the baseline signature indicates an internal winding insulation fault.**

2. Chopped Wave Impulse Voltage Test

In actual operation, a traveling lightning surge often causes a flashover across an adjacent line insulator string. This flashover instantly grounds the line, causing the voltage to collapse to zero almost instantaneously.

- **Test Setup:** A controlled spark gap (e.g., rod gap) is connected in parallel with the transformer terminal. It is configured to flash over and ground the surge wave after a brief delay, typically between **2 μs and 6 μs** from the start of the impulse.
- **Severe Voltage Stress Gradient:** The sudden flashover creates an extremely steep falling edge with a massive rate of voltage collapse (high negative dv/dt). This rapid voltage drop generates extreme dielectric stress across the inter-turn insulation of the coils closest to the line terminal.
- **Fault Detection:** Because the chopped wave's rapid collapse causes chaotic high-frequency oscillations, real-time waveform analysis is difficult. Instead, faults are identified by applying a final full-wave impulse immediately after the chopped wave test. If the neutral current oscillogram from this final full wave matches the initial baseline signature perfectly, the insulation survived the severe dv/dt stress.

Question 5(ii)

Problem Statement:
Write Short notes on: Development of voltage stress along the winding of a three phase transformer for input impulse and RMS voltage. And also write the measure to be taken to withstand it.

Thought Process

Key physical concepts: 1. RMS Voltage: Dominated by winding resistance and inductance. Yields linear voltage distribution. 2. Impulse Voltage: High-frequency transient makes inductance block current. Distribution governed by series/shunt capacitance network. Yields highly exponential distribution, stressing the line-end turns. 3. Measures: Electrostatic shielding, graded insulation.

Answer:
The voltage distribution and resulting dielectric stress along the length of a transformer winding depend entirely on the frequency of the applied voltage excitation.

1. Voltage Stress Under Normal Power Frequency (RMS) Voltage

At standard operational frequencies (50 Hz), the capacitive reactance of the transformer winding ($X_C = 1/\omega C$) is exceedingly high, effectively acting as an open circuit. Capacitive leakage is negligible. The voltage distribution is governed purely by the uniformly distributed series resistance (R) and leakage inductance (L).

Consequently, the steady-state RMS voltage decreases **completely linearly** from the high-voltage line terminal (100% potential) down to the grounded neutral terminal (0% potential). The voltage gradient ($\frac{dv}{dx}$) is constant, meaning every individual turn experiences the exact same, low dielectric stress.

2. Voltage Stress Under Transient Impulse Voltage

An impulse surge features a microsecond-scale steep front, acting as an extremely high-frequency transient (in the MHz range). At these frequencies, the inductive reactance ($X_L = \omega L$) becomes massive, acting as a choke. The initial voltage distribution is dictated entirely by the winding's capacitance network: the series inter-turn capacitance (C_s) and the shunt capacitance to ground (C_g).

As the surge enters the winding, charging current continuously bleeds off to ground through C_g, causing the voltage traveling through C_s to decay exponentially. The initial voltage distribution factor $\alpha = \sqrt{C_g/C_s}$ typically ranges from 5 to 30. This high α causes a highly non-linear distribution where up to **70% to 80% of the total surge voltage drops across the first 10% of the turns** nearest the line terminal. This creates massive dielectric stress on the end-turn insulation.

3. Remedial Measures to Withstand Impulse Stresses

To protect against these destructive capacitive gradients, designers implement two primary strategies:

1. **Graded and Reinforced Insulation:** The inter-turn paper insulation is physically thickened (reinforced) on the coils closest to the line terminals to withstand the severe transient stress.
2. **Electrostatic Shielding (Static Rings):** Metallic static shield rings are installed at the line ends of the windings. These shields artificially increase the series capacitance (C_s) between adjacent coils. By increasing C_s relative to C_g (lowering the α factor), the initial capacitive voltage distribution is forced to become much more linear, significantly reducing the concentrated stress on the first few turns.